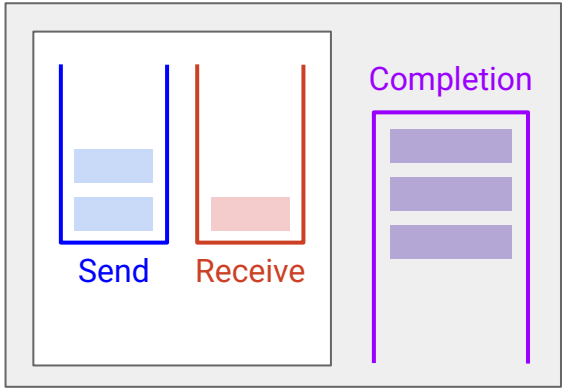



Lecture 20 (Datacenters 4)

# Host Networking, RDMA

**CS 168, Spring 2026 @ UC Berkeley**

Slides credit: Sylvia Ratnasamy, Rob Shakir, Peyrin Kao, Nandita Dukkipati

# What is Host Networking?

Lecture 20, CS 168, Spring 2026

## Optimizations at the Host

- **What is Host Networking?**
- Shared Memory in User Space
- Offloading

Brief History of Offloading

- Epoch 0: No Offloading
- Epoch 1: Stateless Operations
- Epoch 2: Stateful Operations
- Epoch 3: Protocols (RDMA)
- RDMA Example

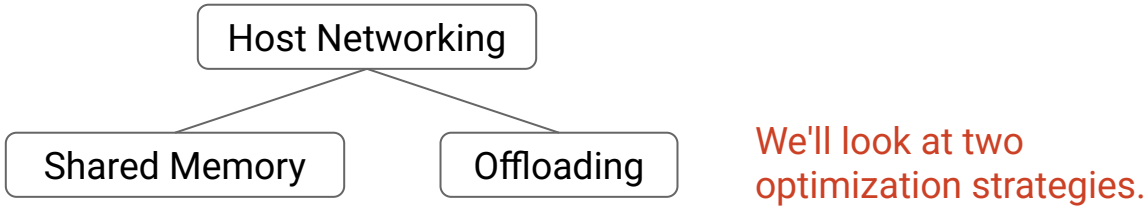
# The New Bottleneck: Hosts

Historically, the network has been the bottleneck. *(Example: Congestion control.)*

In modern datacenters, hosts are the new bottleneck.

- Extremely high performance demands.
- Problem 1: CPUs can't keep up.
  - CPU working on network protocols → less resources for the application.
- Problem 2: Protocols can't keep up.
  - Traditional networking stack (TCP/IP) are insufficient.

**Host networking**: Optimizations **at the host** to meet modern performance demands.



# Optimization: Shared Memory in User Space

Lecture 20, CS 168, Spring 2026

## Optimizations at the Host

- What is Host Networking?
- **Shared Memory in User Space**
- Offloading

Brief History of Offloading

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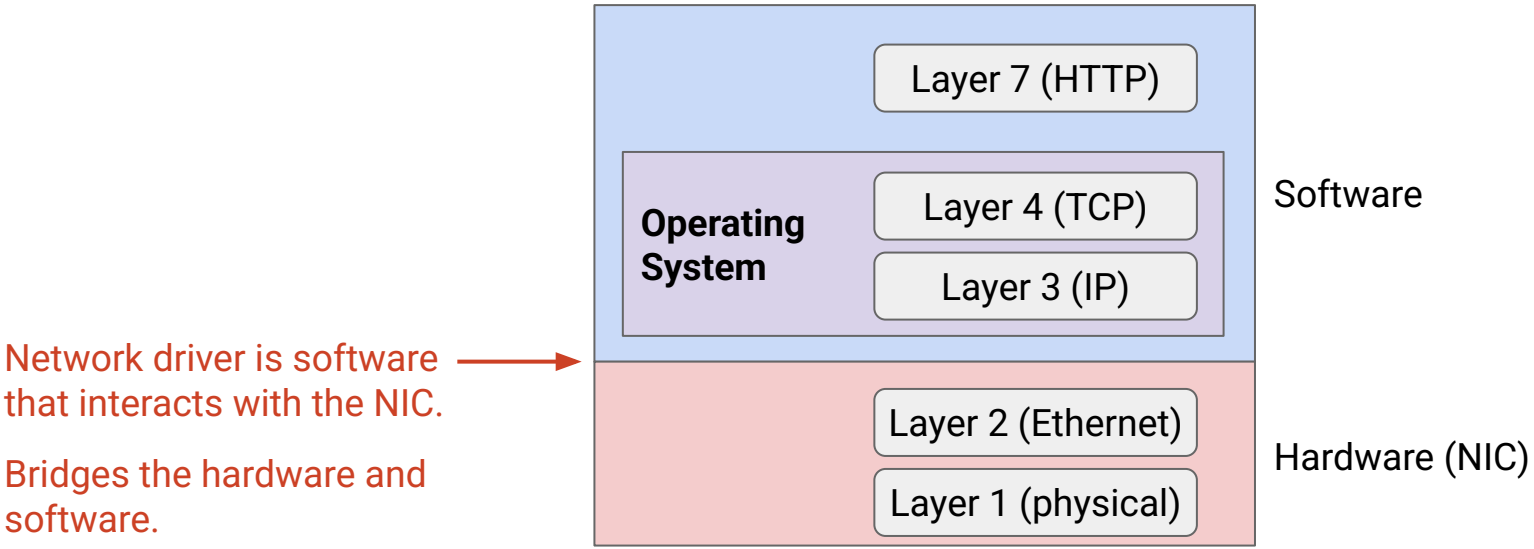
# Recall: Layers in the End Host

Layers 1 and 2 are implemented in hardware, on the network interface card (NIC).

Layers 3 and 4 are implemented in software, in the operating system.

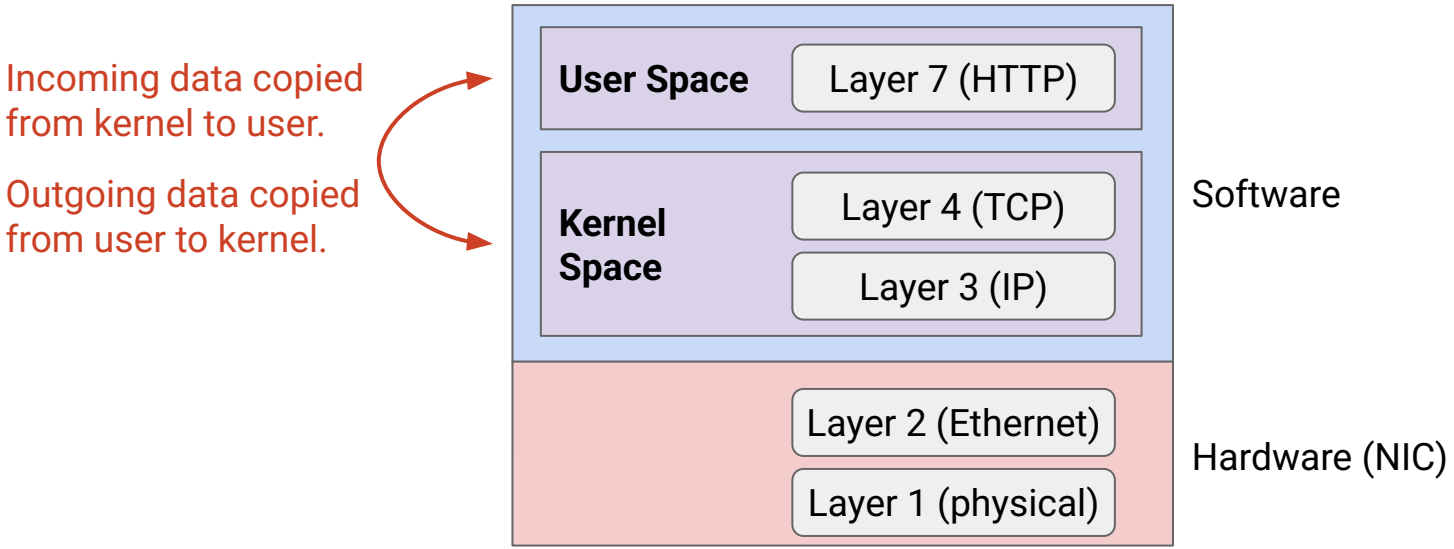
Layer 7 is the applications running in software.



# Problem: Kernel Space and User Space

The OS runs in **kernel space**, and applications run in **user space**.

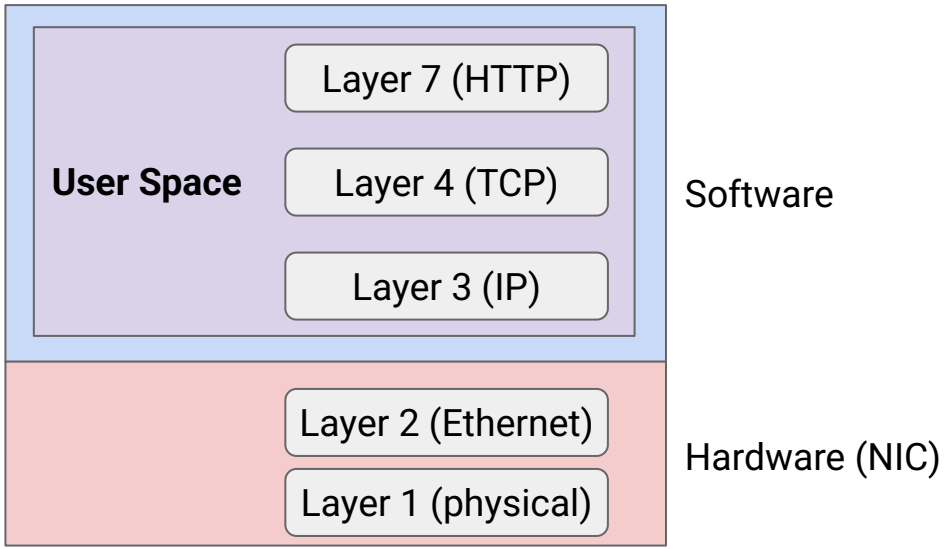
- The two address spaces are isolated from each other. *(Virtual memory from CS 61C.)*
- Problem: Data is constantly being copied between user/kernel space.
- Problem: Programming in the kernel is hard.



# Solution: Networking Stack in User Space

Solution: Packet processing happens in user space.

- Application and network protocols access shared memory.
- No need to copy data back-and-forth!
- Easier to program (e.g. new protocols) in the user space!



# Optimization: Offloading

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## Optimizations at the Host

- What is Host Networking?
- Shared Memory in User Space
- **Offloading**

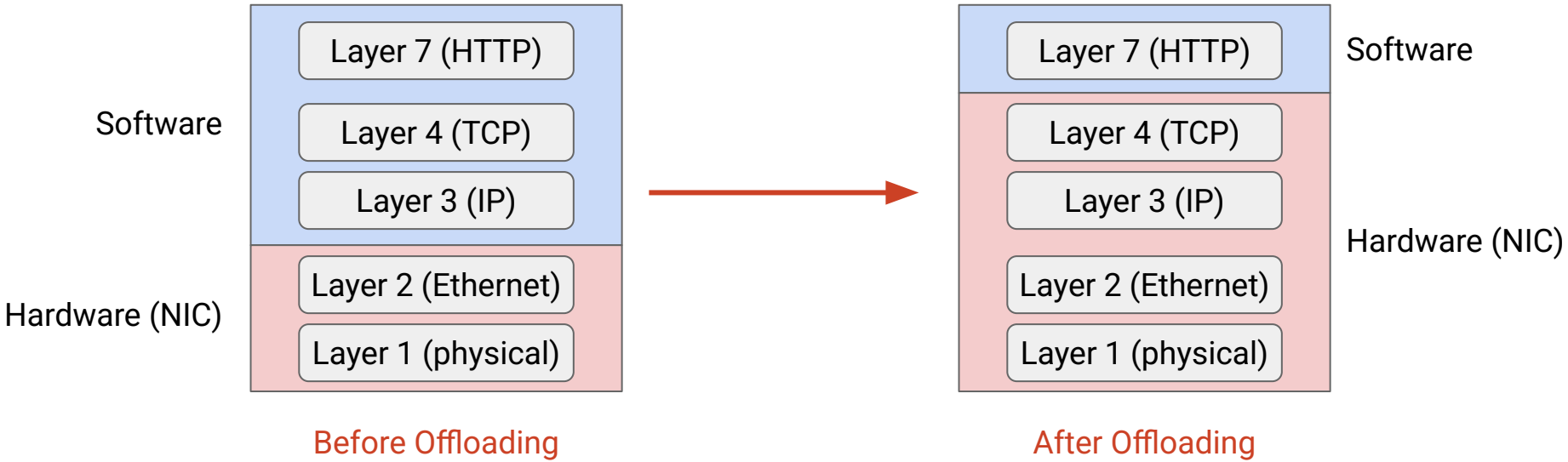
Brief History of Offloading

- Epoch 0: No Offloading
- Epoch 1: Stateless Operations
- Epoch 2: Stateful Operations
- Epoch 3: Protocols (RDMA)
- RDMA Example

# Optimization: Offloading

Problem: CPUs can't keep up with modern performance demands.

Solution: **Offloading**. Process packets in hardware instead of software.



## Benefits of Offloading

Frees up CPU cycles for applications.

- Network processing and applications no longer contend for CPU.

Efficiency.

- Specialized processing in hardware can be more efficient.
- Power savings.

Predictability.

- CPU has unpredictable scheduling delays.
- Hardware gives us more consistent latencies.

# Offloading, Epoch 0: No Offloading

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Optimizations at the Host

- What is Host Networking?
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- Offloading

**Brief History of Offloading**

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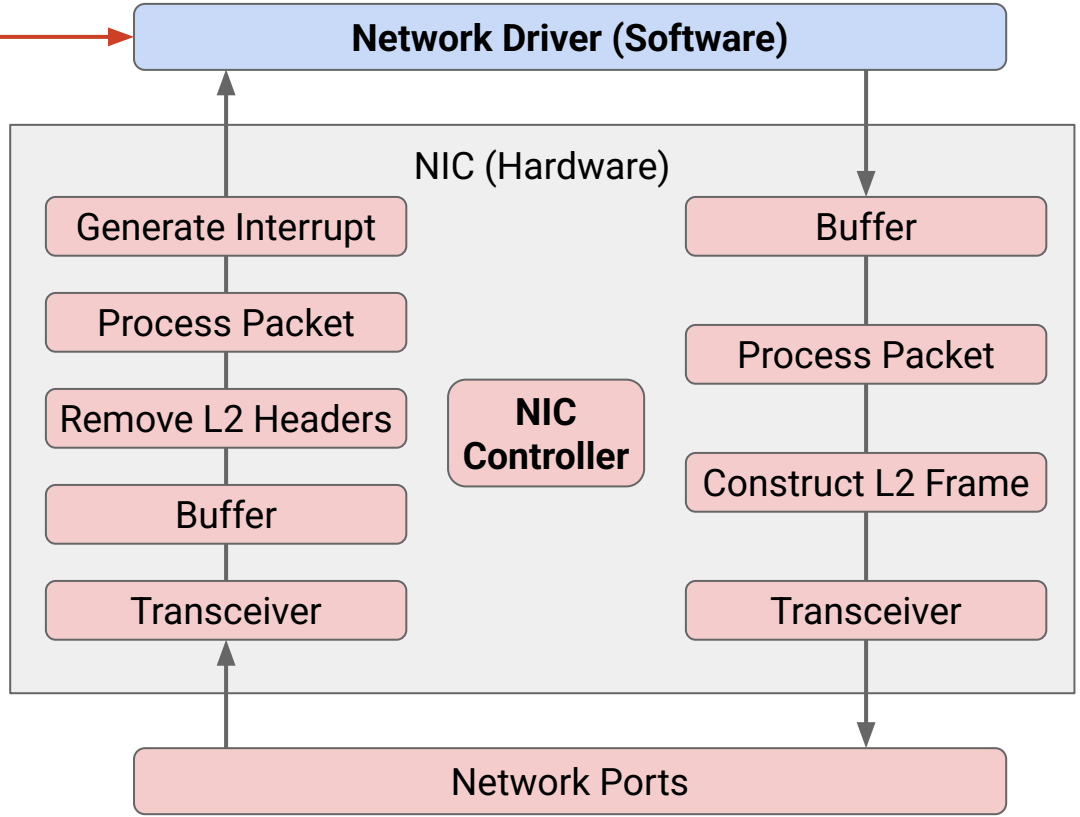
# NIC Before Offloading

Before any offloading, the NIC is a "doormat" for packets, with minimal processing.

Network driver is software that interacts with the NIC.

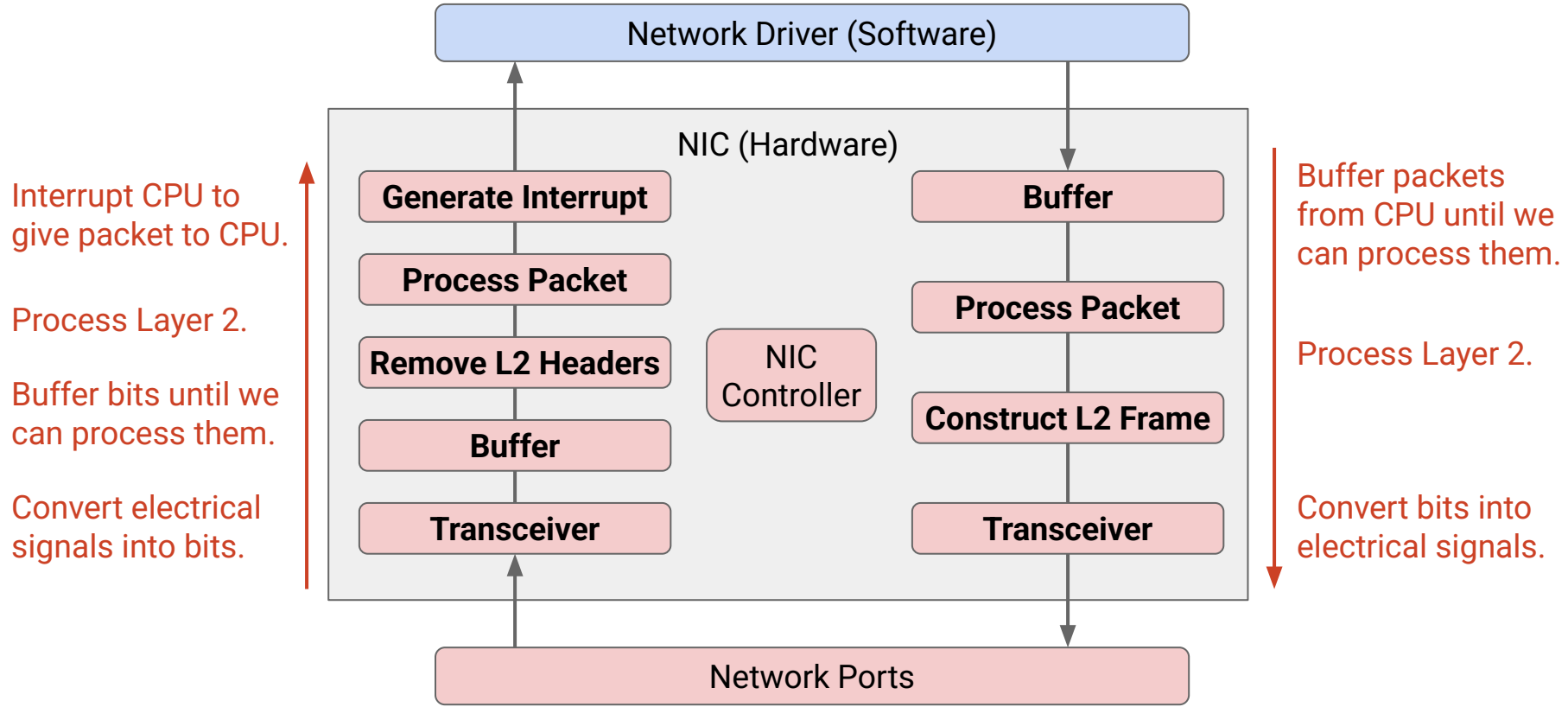
Bridges the hardware and software.




NIC controller manages operation on the card.

# NIC Before Offloading

The NIC passes incoming packets up to CPU, and sends out packets from CPU.



# Offloading, Epoch 1: Stateless Operations

Lecture 20, CS 168, Spring 2026

Optimizations at the Host

- What is Host Networking?
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**Brief History of Offloading**

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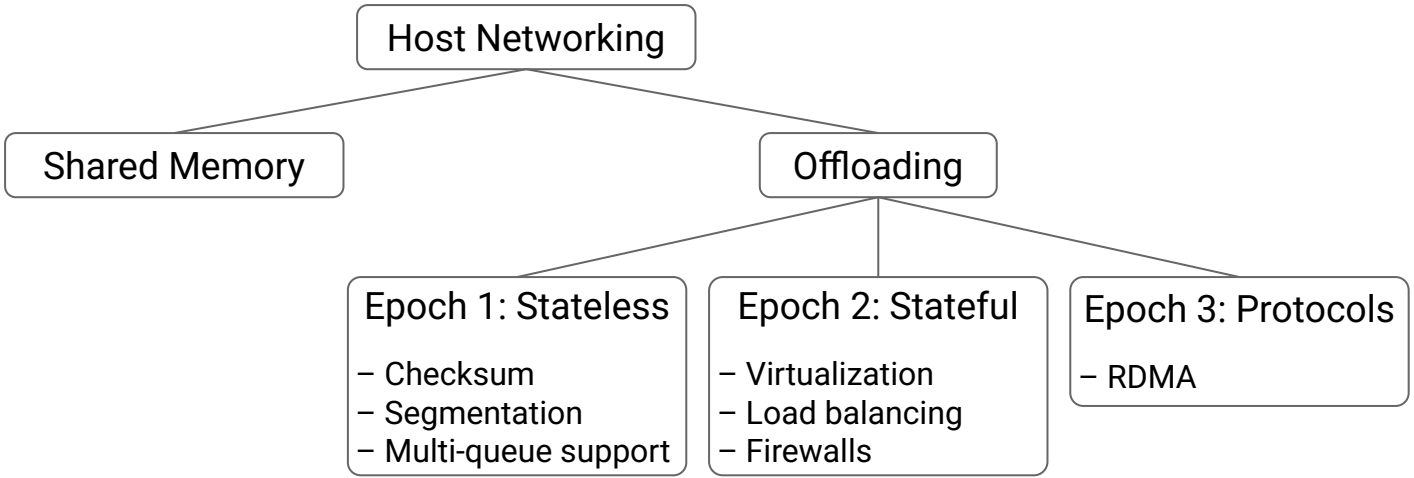
- RDMA Example

# Brief History of Offloading

Over time, increasingly complicated operations were offloaded to the NIC.

- Epoch 1: Simple stateless operations.
- Epoch 2: Stateful operations.
- Epoch 3: Protocols like RDMA.



# Epoch 1 Offloads (1/3): Checksum

The NIC can *set* Layer 4 checksum for *outgoing* packets.

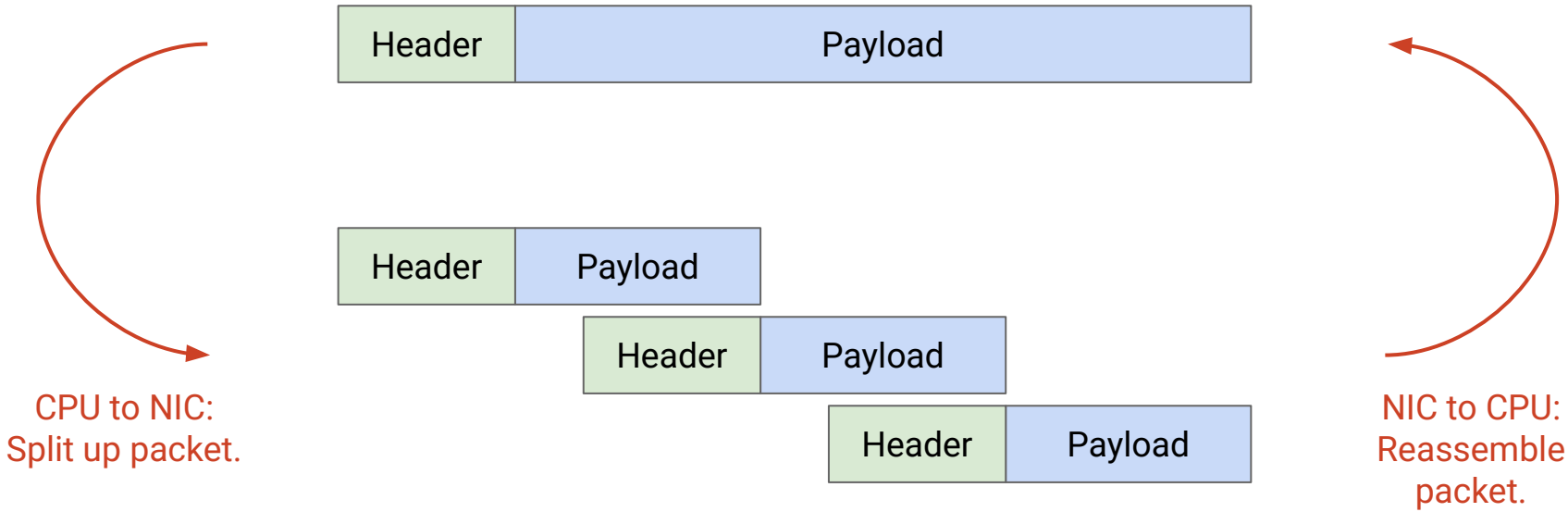
The NIC can *verify* Layer 4 checksum for *incoming* packets.

CPU tells the NIC where to read/write the checksum.

# Epoch 1 Offloads (2/3): Segmentation

CPU can offload segmentation work (splitting and reassembling packets) to the NIC.

- CPU handles a few large packets.
- NIC handles lots of small packets.



## Epoch 1 Offloads (2/3): Segmentation

Segmentation comes with trade-offs.

With segmentation:

- Less work for CPU to do.
- CPU hands large packets to NIC. Traffic is more bursty.

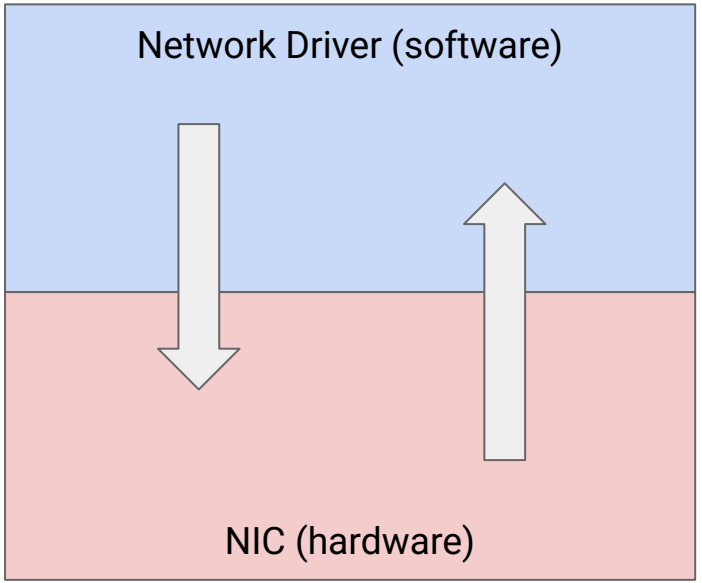
Without segmentation:

- More work for CPU to do.
- CPU hands small packets to NIC. Traffic is smoother.

# Epoch 1 Offloads (3/3): Multi-Queue Support

Without offloading: NIC has 1 transmit queue and 1 receive queue.

Network driver is responsible for isolation and load balancing in a multi-processor system (multiple CPUs).



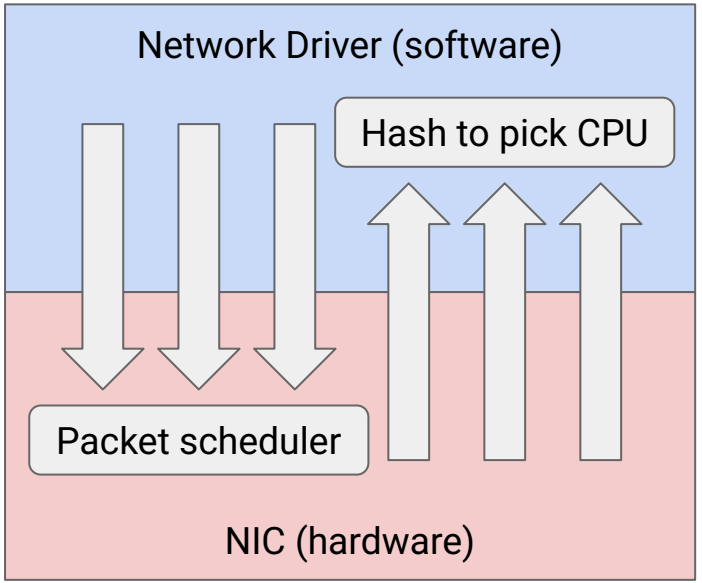
# Epoch 1 Offloads (3/3): Multi-Queue Support

Idea: Multiple transmit queues.

- Packet scheduler decides which queue to send from next.

Idea: Multiple receive queues.

- Hash packets (like ECMP, equal-cost multi-path routing) to decide which CPU handles the packet.



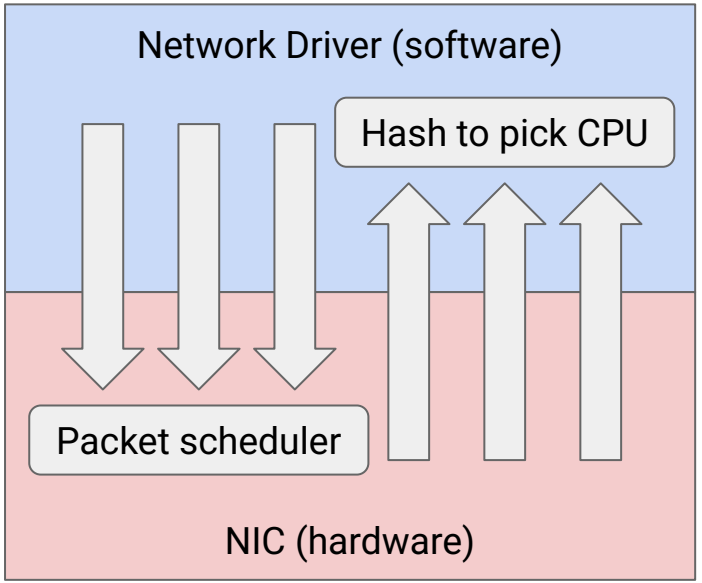
# Epoch 1 Offloads (3/3): Multi-Queue Support

Benefits:

- Ensures isolation: Specific flows can use a dedicated queue.
- Ensure load-balancing: Heavy traffic in one queue doesn't affect others.
- Can prioritize specific flows.

Challenges:

- Must ensure packets in the same flow use the same queue, to avoid out-of-order packets (which slows down TCP).



# Offloading, Epoch 2: Stateful Operations

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Optimizations at the Host

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**Brief History of Offloading**

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- Epoch 3: Protocols (RDMA)
- RDMA Example

## Epoch 2: Stateful Offloads with Match-Action Tables

Stateful operations to offload:

- **Virtualization**: Virtual switch forwarding packets to/from VMs.
- **Bandwidth metering**: If user sends too many packets, drop the excess.
- **Firewalls**: Drop packets from malicious users.

Behavior specified by match-action tables (like OpenFlow).

| **Match** | **Action** |
|---|---|
| Source IP = 10.1.8.9 | Add encapsulation header, forward. |
| 5-tuple matches<br>(10.1.2.3, 50000, TCP, 24.1.3.0, 80) | Allow 100 packets per minute.<br>Drop excess. |
| Source IP = 76.124.1.2 | Malicious source. Drop. |

# Offloading, Epoch 3: Protocols (RDMA)

Lecture 20, CS 168, Spring 2026

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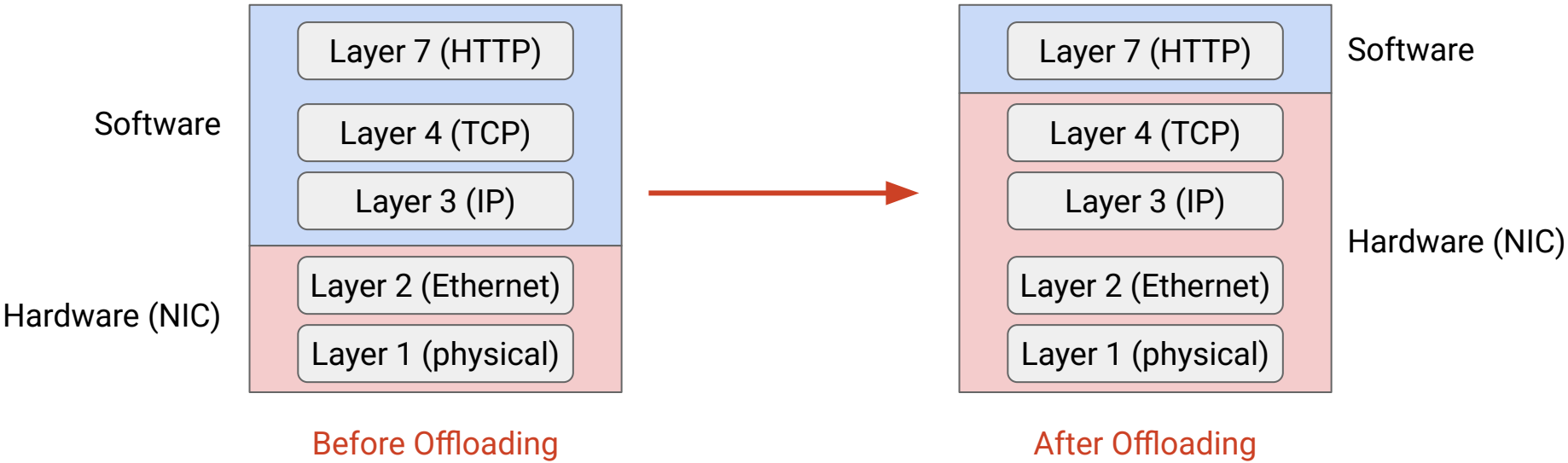
# Epoch 3: Protocols

Goal: Offload entire protocols, like TCP, into hardware.

- Active area of research!
- Instead of implementing TCP in hardware, design custom protocols for hardware.



# RDMA: Remote Direct Memory Access

Transferring a file with traditional TCP/IP networking stack:

- Sender CPU reads file from memory.
- Sender CPU sends out packets.
- Recipient CPU processes packets.
- Recipient CPU writes file to memory.


Without RDMA: CPU involved in data transfer.

# RDMA: Remote Direct Memory Access

**RDMA** (Remote Direct Memory Access):

- Abstraction: A can directly access memory in B's computer.
- Goal: Minimal CPU involvement in the data transfer.

Transferring a file with RDMA:

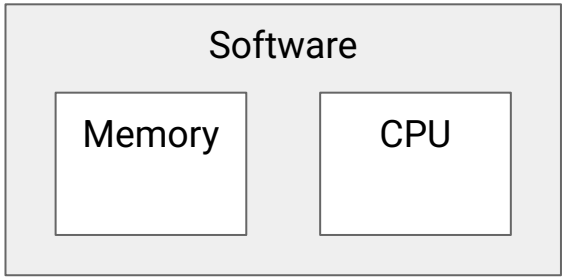
- CPUs set up the transfer and get out of the way.
- Sender NIC reads memory and sends out data.
- Recipient NIC receives data and writes to memory.


RDMA: CPU is minimally involved in transfer!

# Abstraction: Queue Pairs

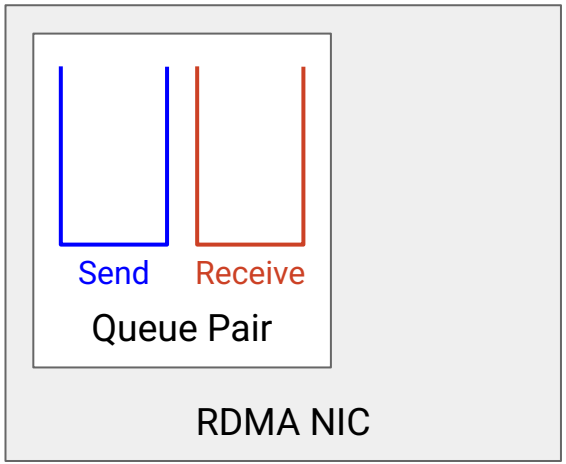
TCP used bytestreams, but RDMA uses the **queue pair** abstraction between the application and NIC.

- **Send queue** has all pending jobs, where I'm sending data to someone else.
- **Receive queue** has all pending jobs, where someone else is sending data to me.




A single NIC can have multiple queue pairs.

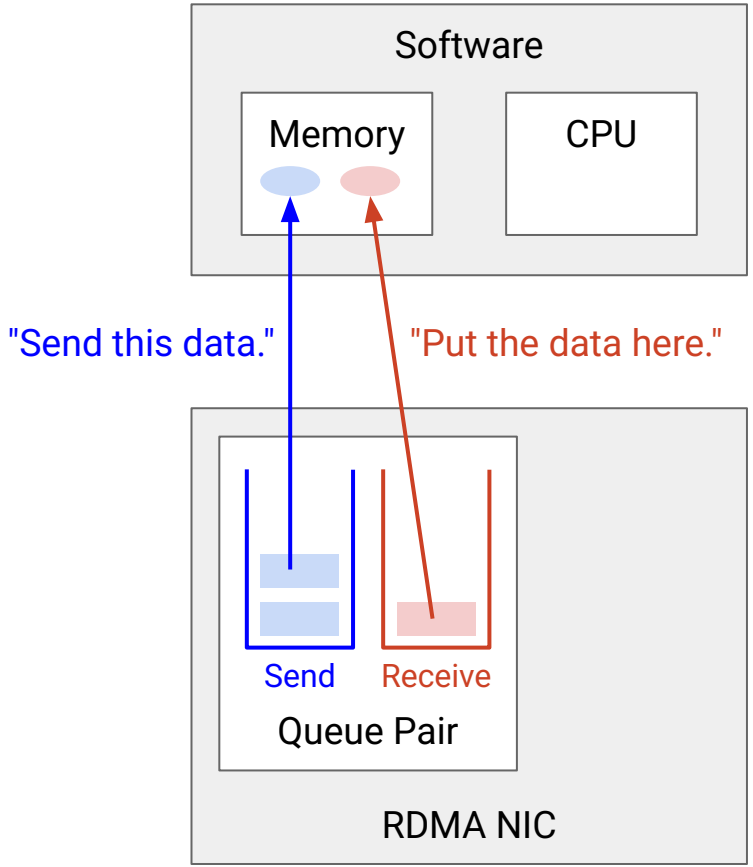
- Each queue pair has a different service guarantee. (Reliable? Ordered?)
- Reliable, in-order queue pair is most similar to TCP.



# Abstraction: Work Queue Elements

CPU schedules transfers by adding **Work Queue Elements (WQEs)** to the queues.

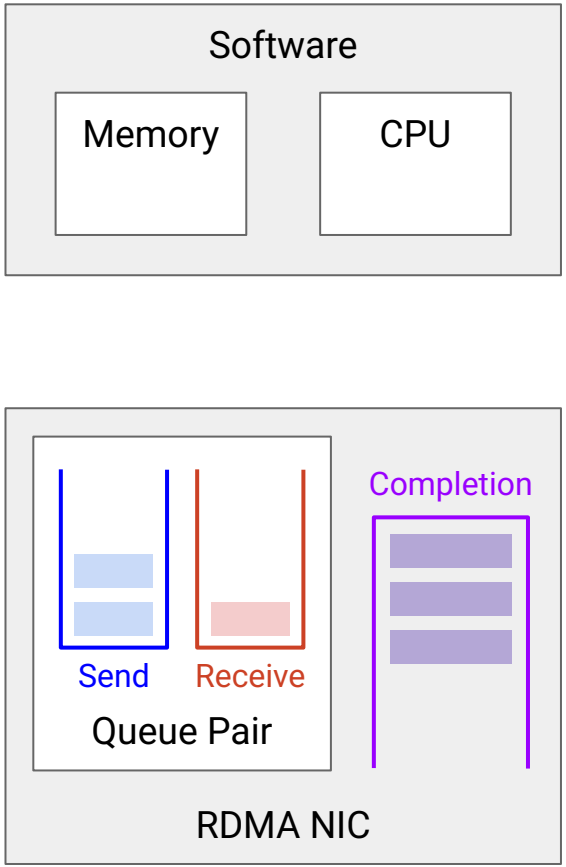
- Struct with instructions about the transfer.
  - Contains pointers to where in memory to read/write data.
- Gives the NIC an application-level view of the transfer, e.g. where the file starts/ends.
  - Contrast with TCP, which just sees a bytestream.



# Abstraction: Asynchronous Scheduling, Completion Queue

RDMA is asynchronous.

- CPU schedules jobs in the queue pair.
- NIC processes jobs in the queue.
- When job completes, NIC adds a **Completion Queue Element (CQE)** to the Completion Queue, describing what happened (e.g. success or error).
- Application reads CQE to see what happened with the job.



# RDMA Example

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## RDMA Example

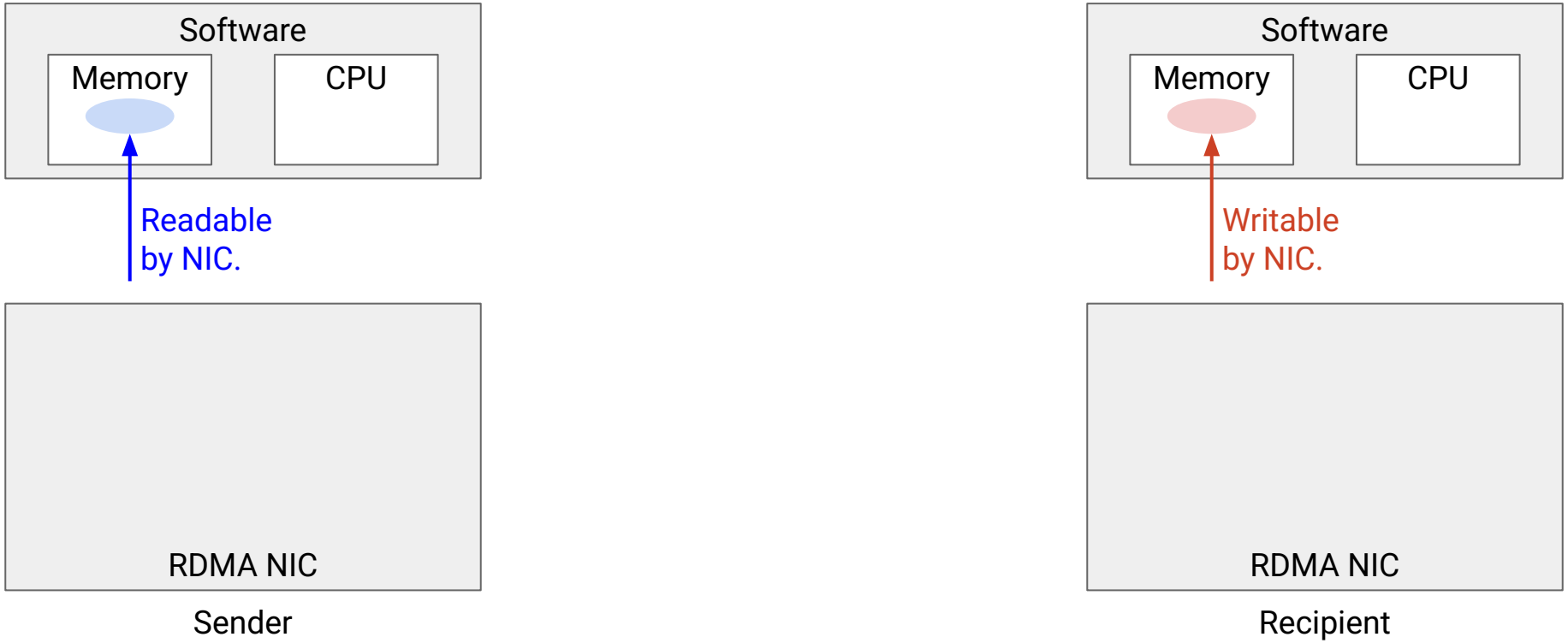
There are many different RDMA operations:

- RDMA Write.
- RDMA Read.
- RDMA Atomic. Perform an operation in remote host's memory.
- RDMA Write with Immediate. Send additional value along with writing data.

Let's look at an RDMA Send/Receive.

# Steps of RDMA (1/6)

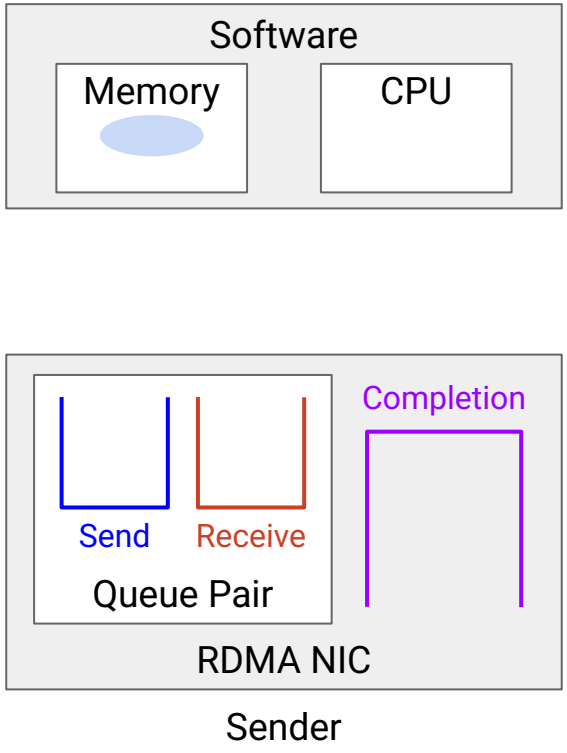
1. Each server designates some memory to be accessible by NIC for RDMA transfers.

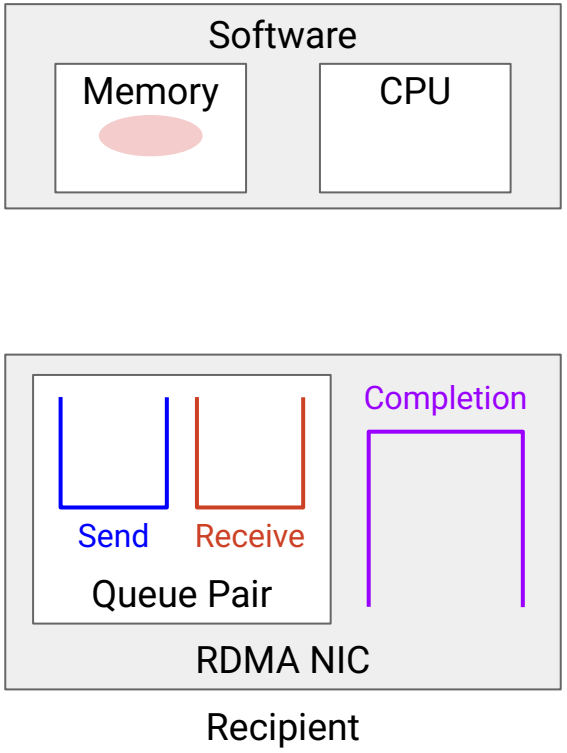


# Steps of RDMA (2/6)

2. Each server sets up queues.

This can be done out-of-band, e.g. use TCP to coordinate between servers.

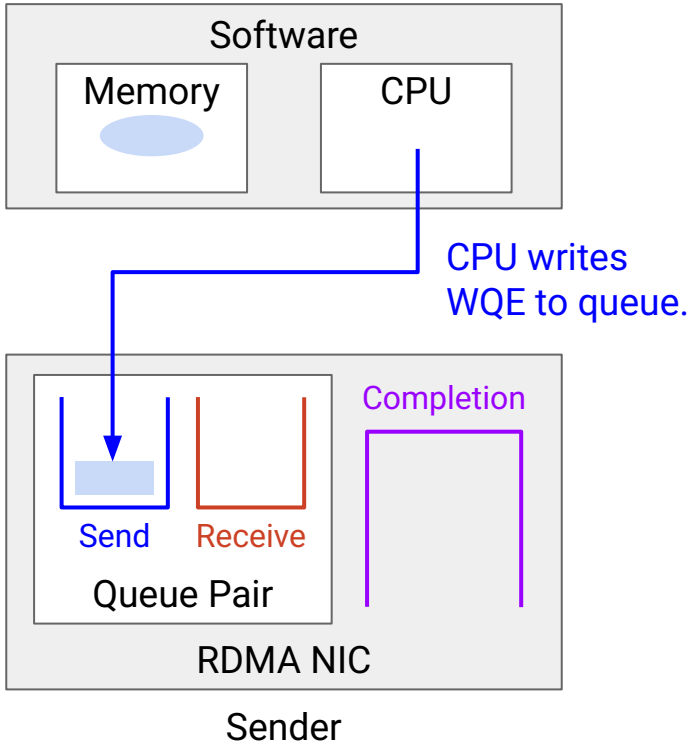


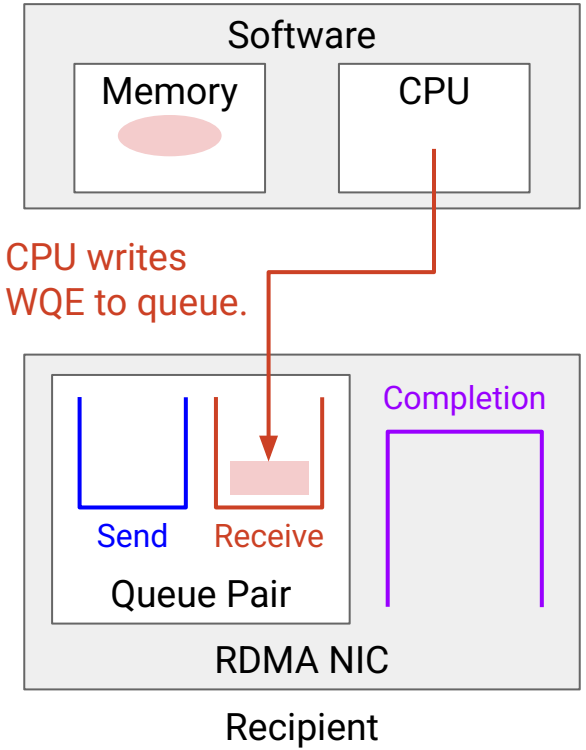



# Steps of RDMA (3/6)

3. Servers set up Work Queue Entries (WQEs) in the queues.

WQE contains pointer to buffer in memory.



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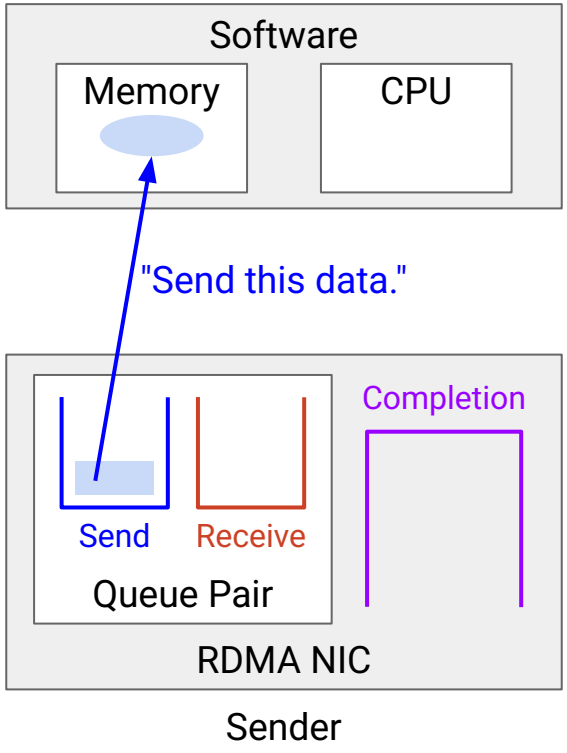
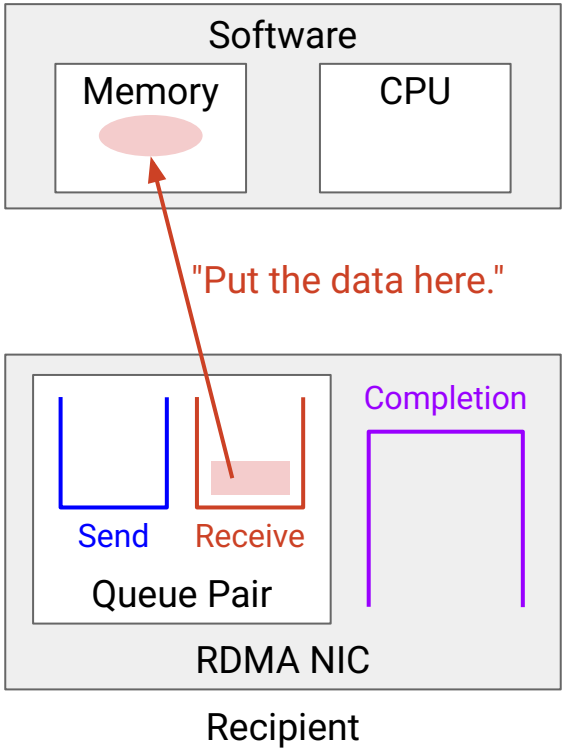
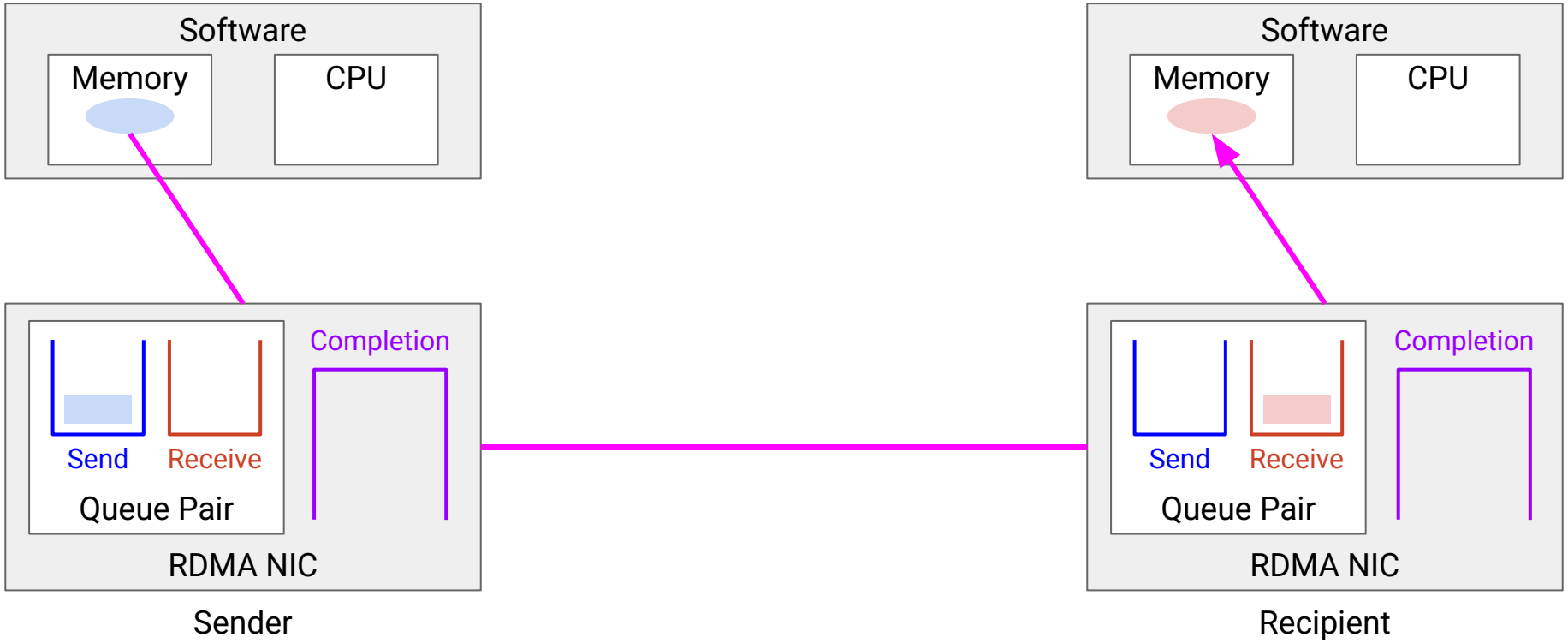


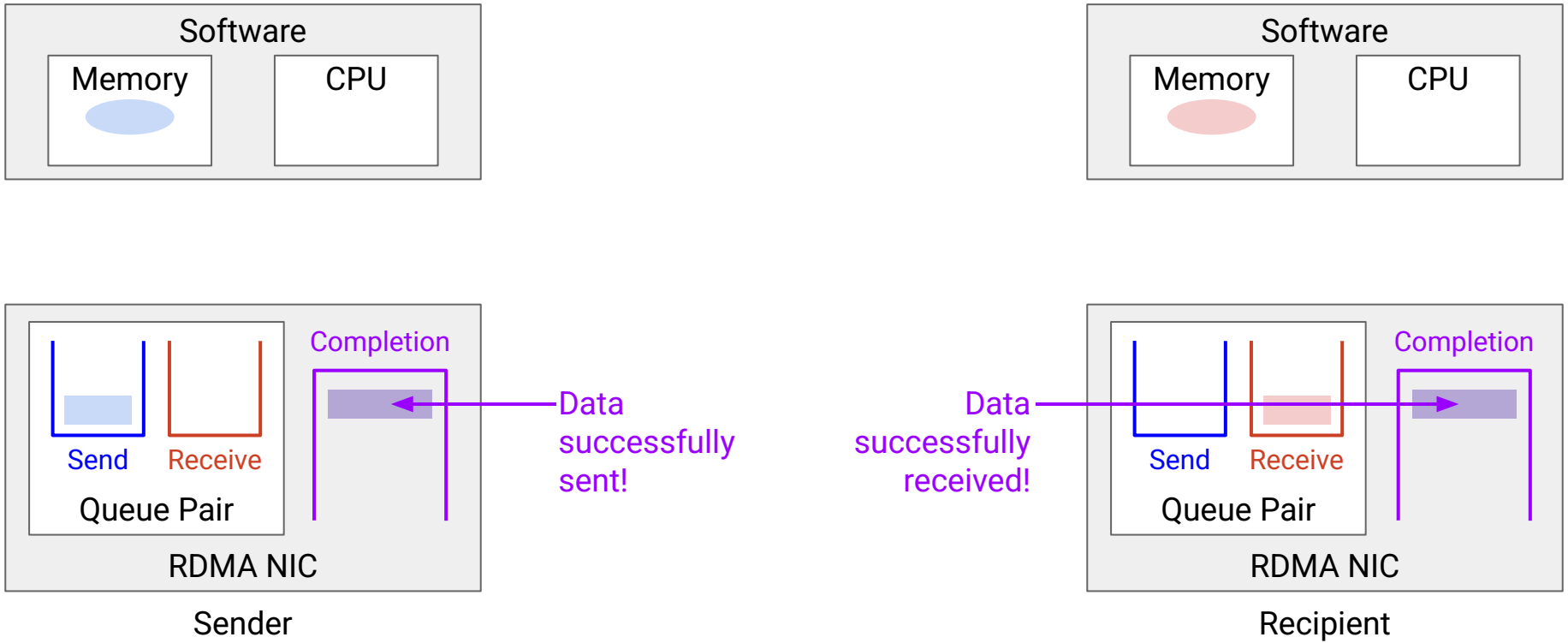
# Steps of RDMA (4/6)

4. NIC transfers data between memory, without CPU involvement!



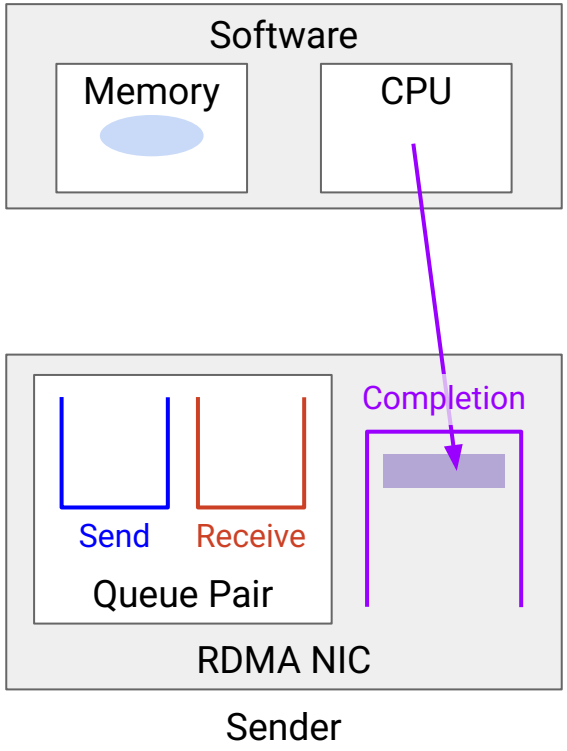
# Steps of RDMA (5/6)

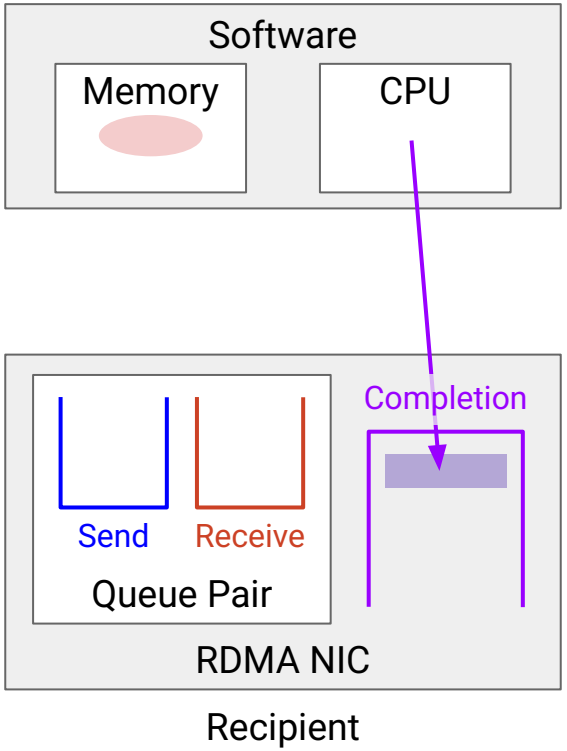
5. NICs generate Completion Queue Entries (CQEs), and delete WQEs.



# Steps of RDMA (6/6)

6. Applications read CQE to understand what happened to the transfer.



# RDMA: Pros and Cons

Pros:

- High-performance data transfer.
- Frees up CPU for applications.

Limitations:

- More complex. Requires specialized hardware and software.
  - RDMA replaces TCP/IP, so features like reliability must be implemented in hardware.
- Works best when servers are nearby, e.g. in same datacenter.
  - If servers are far away, dominant delay is the network. RDMA savings are insignificant.

# RDMA: Applications

RDMA useful for high-performance computing.

- Scientific research, financial modeling, weather forecasting, etc.
- Cloud computing, e.g. migrating large VM from one physical server to another.
- AI/ML training.

Goals:

- Free up CPU for applications.
- Low and predictable latency for data transfers.

# RDMA: Implementation

How do you handle losses, reordering, congestion, etc.?

Two broad options:

- Build features into the datacenter network, e.g. at the switches. (*Nvidia's InfiniBand.*)
- Build features in the NIC, under the queue pair abstraction. (*Google.*)

In both cases, applications and OS both see the abstraction of reliable delivery.